INCH-POUND

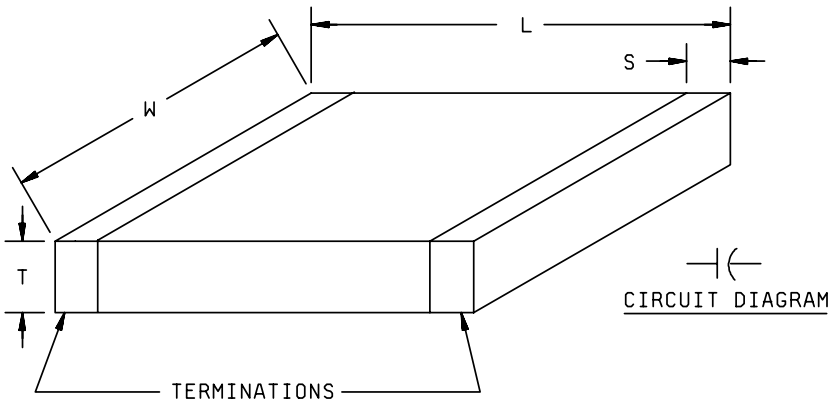
MIL-PRF-123/10C
1 September 2005
SUPERSEDING
MIL-PRF-123/10B
18 June 2001

PERFORMANCE SPECIFICATION SHEET

CAPACITORS, FIXED, CERAMIC DIELECTRIC, (TEMPERATURE STABLE AND GENERAL PURPOSE), HIGH RELIABILITY, NONLEADED, STYLE CKS51

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-PRF-123.



Dimensions				
L ± .015	W ± .015	T		S ± .010
		Min	Max	
.080	.050	.020	.055	.020

Inches	mm
.010	0.25
.015	0.38
.020	0.51
.050	1.27
.055	1.40
.080	2.03

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Dimensions and tolerances are for bare chips. For solder coated terminations, add .020 inch to the positive length tolerance and .015 inch to the positive width and thickness tolerances.

FIGURE 1. Style CKS51 capacitors.

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REQUIREMENTS:

Dimensions and configuration: See figure 1.

Case type: Multi-layer, unencapsulated, monolithic ceramic.

Termination: M, G, S, or W in accordance with MIL-PRF-123.

Capacitance value: See table I.

Capacitance tolerance: See table I.

Operating temperature: -55°C to +125°C.

Rated voltage: 50 V dc.

Marking: In accordance with MIL-PRF-123.

TABLE I. Style CKS51 characteristics.

Part or Identifying Number (PIN) 1/	Capacitance (pF)	Capacitance tolerance	Voltage-temperature limits	Rated voltage (volts)
M123A10BPB1R0--	1.0	C, D	BP	50
M123A10BPB1R1--	1.1	C, D	BP	50
M123A10BPB1R2--	1.2	C, D	BP	50
M123A10BPB1R3--	1.3	C, D	BP	50
M123A10BPB1R5--	1.5	C, D	BP	50
M123A10BPB1R6--	1.6	C, D	BP	50
M123A10BPB1R8--	1.8	C, D	BP	50
M123A10BPB2R0--	2.0	C, D	BP	50
M123A10BPB2R2--	2.2	C, D	BP	50
M123A10BPB2R4--	2.4	C, D	BP	50
M123A10BPB2R7--	2.7	C, D	BP	50
M123A10BPB3R0--	3.0	C, D	BP	50
M123A10BPB3R3--	3.3	C, D	BP	50
M123A10BPB3R6--	3.6	C, D	BP	50
M123A10BPB3R9--	3.9	C, D	BP	50
M123A10BPB4R3--	4.3	C, D	BP	50
M123A10BPB4R7--	4.7	C, D	BP	50
M123A10BPB5R1--	5.1	C, D	BP	50
M123A10BPB5R6--	5.6	C, D	BP	50
M123A10BPB6R2--	6.2	C, D	BP	50
M123A10BPB6R8--	6.8	C, D	BP	50
M123A10BPB7R5--	7.5	C, D	BP	50
M123A10BPB8R2--	8.2	C, D	BP	50
M123A10BPB9R1--	9.1	C, D	BP	50
M123A10BPB100--	10	C, J, K	BP	50
M123A10BPB110--	11	C, J, K	BP	50
M123A10BPB120--	12	C, J, K	BP	50
M123A10BPB130--	13	C, J, K	BP	50
M123A10BPB150--	15	C, J, K	BP	50
M123A10BPB160--	16	C, J, K	BP	50
M123A10BPB180--	18	C, J, K	BP	50
M123A10BPB200--	20	C, J, K	BP	50

See footnote at end of table.

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* TABLE I. Style CKS51 characteristics - Continued.

PIN 1/	Capacitance (pF)	Capacitance tolerance	Voltage-temperature limits	Rated voltage (volts)
M123A10BPB220--	22	C, J, K	BP	50
M123A10BPB240--	24	C, J, K	BP	50
M123A10BPB270--	27	F, J, K	BP	50
M123A10BPB300--	30	F, J, K	BP	50
M123A10BPB330--	33	F, J, K	BP	50
M123A10BPB360--	36	F, J, K	BP	50
M123A10BPB390--	39	F, J, K	BP	50
M123A10BPB430--	43	F, J, K	BP	50
M123A10BPB470--	47	F, J, K	BP	50
M123A10BPB510--	51	F, J, K	BP	50
M123A10BPB560--	56	F, J, K	BP	50
M123A10BPB620--	62	F, J, K	BP	50
M123A10BPB680--	68	F, J, K	BP	50
M123A10BPB750--	75	F, J, K	BP	50
M123A10BPB820--	82	F, J, K	BP	50
M123A10BPB910--	91	F, J, K	BP	50
M123A10BPB101--	100	F, J, K	BP	50
M123A10BPB111--	110	F, J, K	BP	50
M123A10BPB121--	120	F, J, K	BP	50
M123A10BPB131--	130	F, J, K	BP	50
M123A10BPB151--	150	F, J, K	BP	50
M123A10BPB161--	160	F, J, K	BP	50
M123A10BPB181--	180	F, J, K	BP	50
M123A10BPB201--	200	F, J, K	BP	50
M123A10BPB221--	220	F, J, K	BP	50
M123A10BPB241--	240	F, J, K	BP	50
M123A10BPB271--	270	F, J, K	BP	50
M123A10BPB301--	300	F, J, K	BP	50
M123A10BPB331--	330	F, J, K	BP	50
M123A10BPB361--	360	F, J, K	BP	50
M123A10BPB391--	390	F, J, K	BP	50
M123A10BPB431--	430	F, J, K	BP	50
M123A10BPB471--	470	F, J, K	BP	50
M123A10BPB511--	510	F, J, K	BP	50
M123A10BPB561--	560	F, J, K	BP	50
M123A10BPB621--	620	F, J, K	BP	50
M123A10BPB681--	680	F, J, K	BP	50
M123A10BXB331K-	330	K	BX	50
M123A10BXB391K-	390	K	BX	50
M123A10BXB471K-	470	K	BX	50
M123A10BXB561K-	560	K	BX	50
M123A10BXB681K-	680	K	BX	50
M123A10BXB821K-	820	K	BX	50
M123A10BXB102K-	1,000	K	BX	50
M123A10BXB122K-	1,200	K	BX	50
M123A10BXB152K-	1,500	K	BX	50
M123A10BXB182K-	1,800	K	BX	50
M123A10BXB222K-	2,200	K	BX	50
M123A10BXB272K-	2,700	K	BX	50
M123A10BXB332K-	3,300	K	BX	50
M123A10BXB392K-	3,900	K	BX	50

See footnote at end of table.

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* TABLE I. Style CKS51 characteristics - Continued.

PIN 1/	Capacitance (pF)	Capacitance tolerance	Voltage-temperature limits	Rated voltage (volts)
M123A10BXB472K-	4,700	K	BX	50
M123A10BXB562K-	5,600	K	BX	50
M123A10BXB682K-	6,800	K	BX	50
M123A10BXB822K-	8,200	K	BX	50
M123A10BXB103K-	10,000	K	BX	50
M123A10BXB123K-	12,000	K	BX	50
M123A10BXB153K-	15,000	K	BX	50
M123A10BXB183K-	18,000	K	BX	50

1/ Complete PIN shall include additional letter(s) to indicate capacitance tolerance and termination, if applicable.

Changes from previous issue: The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Navy - EC
Air Force - 19
DLA - CC
NASA - NA

Preparing activity:
DLA - CC

(Project 5910-2005-018)

* NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using ASSIST Online database at http://assist.daps.dla.mil.

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